VIETNAMESE - GERMAN UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE



Customer Segmentation Analysis REPORT

Customer and Segmentation Data Analytics

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List of Tables

2.1 Comparative Analysis Matrix 3

List of Figures

3.1 Đây là chú thích của hình ảnh. 5

Contents

1 Introduction & Problem Statement **1**

1.1 Context 1

1.1.1 Main issue 1

2 Theoretical Framework & Industry Benchmark Case Studies **2**

2.1 Context 2

2.1.1 Main issue 3

3 Datawarehose Design **5**

3.1 Context 6

3.1.1 Main issue 6

4 Power BI **7**

4.1 Context 7

4.1.1 Main issue 7

Chapter 1

Introduction & Problem Statement

1.1 Context

In retail and e-commerce enterprise systems, understanding customer purchasing patterns is vital for sustainable growth. Analysis of initial transactional data at CompanyX revealed four core business pain points:

- **High Single-Purchase Churn:** 62.9% of customers make only one order and never return. Only 29.5% make a second order, and a minimal 7.4% complete 3 to 5 orders.
- **Basket Size Limitations:** A significant portion of transactions involve minimal item counts (2,492 single-item purchases vs. 4,480 two-item purchases), indicating low initial basket size expansion.
- **Cross-Selling Inefficiencies:** Over 9,000 B2C customers purchased products in the "Bikes" category but never bought complementary items from "Accessories" or "Clothing".
- **Customer Inactivity:** A high proportion of customer profiles remain inactive for extended durations ($>$ 180 days), reducing total Customer Lifetime Value (CLV) and inflating Customer Acquisition Costs (CPA).

1.1.1 Main issue

Describe the key problem statements, current bottlenecks, and target solutions here.

Chapter 2

Theoretical Framework & Industry Benchmark Case Studies

2.1 Case Study Summary

To formulate an effective solution for CompanyX, three industry benchmarks utilizing distinct segmentation methodologies were evaluated:

- **Black Diamond (RFM Segmentation):** Source: Lexer (2025). Specializes in high-end technical climbing and outdoor equipment, transitioning from wholesale to Direct-to-Consumer (D2C) online. Faced with messy data across disparate systems, mass email marketing, and high customer churn, they established a Single Customer View (SCV) and applied Recency, Frequency, and Monetary (RFM) segmentation. Personalized email campaigns targeting inactive ($>$ 180 days) and lapsed ($>$ 1 year) customers yielded a +1,101% revenue per email increase, a 50% reduction in CPA, and a 2x Return on Advertising Spend (ROAS).

- **Zappos (K-Means Clustering):** Source: Towards AI (2020). An e-commerce platform for footwear and apparel. Facing uniform marketing inefficiencies, they deployed unsupervised K-Means clustering on Annual Income and Spending Score data. Using the Elbow Method, they identified 5 distinct customer groups (Target/VIPs, Savers, Spenders, Budget-conscious, and Standard), enabling tailored marketing campaigns and optimized outreach.

- **DANA Indonesia (Data Warehouse + RFM + K-Means):** Source: Bima Putra Pratama / Alibaba Cloud (2020). *Corporate Profile:* Financial Technology (Fintech) digital wallet operating a B2C model. *Business Pain Points:* Simple heuristic rules failed due to highly skewed transactional data, and they lacked an automated data warehouse architec-

ture to daily ingest, standardize, score, and store customer segments. *Solution:* Built an end-to-end customer segmentation pipeline utilizing Alibaba Cloud's MaxCompute (Data Warehouse) and Machine Learning Platform for AI (PAI), applying log transformations, K-Means clustering ($k = 5$), and daily automated batch serving. *Results:* Enabled seamless tracking of historical behavioral shifts over time and allowed marketing teams to query user segments directly via SQL and dashboards.

2.2 Comparative Analysis Matrix

Table 2.1: Comparative Analysis Matrix

Methodology	**Nature**	**Complexity**	**Key Advantages**
RFM Segmentation	Rule-based classification (Recency, Frequency, Monetary).	Low, calculated using simple SQL/BI logic, no ML required.	Easy to implement, low cost, suitable for re-engaging direct marketing campaigns.
K-Means Clustering	Unsupervised machine learning (Unsupervised ML) using features like Annual Income and Spending Score to find natural customer groupings.	Medium, requires applying the Elbow Method to find $k = 5$ clusters and scaling feature data before processing.	Discovers hidden behavioral patterns based on income and spending habits (e.g., Target VIPs vs. High Spenders).
Data Warehouse + RFM + K-Means	Integrates RFM segmentation with K-Means clustering on a Data Warehouse platform.	High, requires skewed data transformation pipelines (log transform/standardization) and daily automated snapshot saving.	Automated segmentation for millions of users, allowing easy querying of behavioral shifts over time.

2.3 Problem Statement

Currently, CompanyX lacks a comprehensive, integrated customer analytics framework to track purchasing patterns and profile customer behaviors. Based on empirical transactional analysis, the primary strategic goal is to formulate a data-driven solution addressing three critical

challenges:

- **Customer Retention & Churn Mitigation:** How can CompanyX re-engage inactive customers and effectively retain the 62.9% majority of one-time buyers who drop off after their initial order?
- **Cross-Selling Optimization:** How can we eliminate cross-selling inefficiencies and drive basket expansion for over 9,000 B2C customers who purchased bikes but never bought complementary accessories or apparel?
- **Data-Driven Segmentation Framework:** How can we establish a scalable Data Warehouse and machine learning pipeline leveraging K-Means Clustering to automatically segment CompanyX's customer base into distinct, actionable value tiers and behavioral profiles?

Chapter 3

Datawarehose Design

We propose our data warehouse design as below:

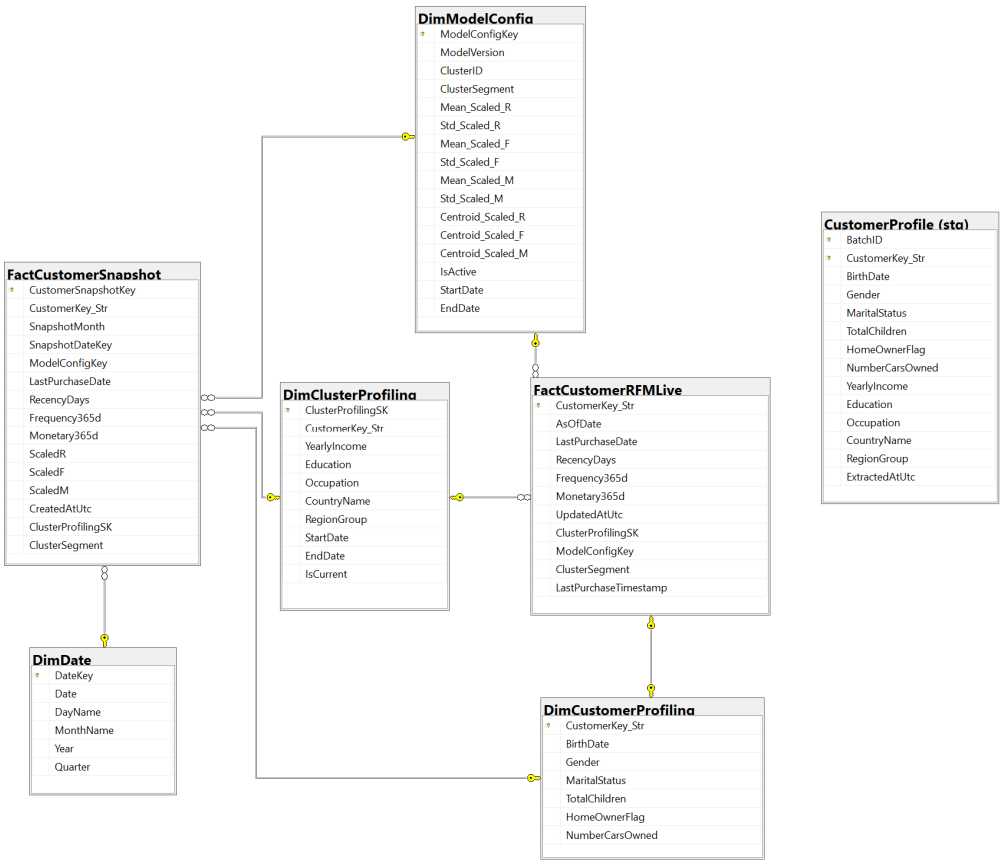


Figure 3.1: Đây là chú thích của hình ảnh.

At a first glance, we can observe that the blueprint follows a hybrid constellation / star-galaxy schema which consists of two fact tables, namely FactCustomerSnapshot, FactCustomerRFMLive; four dimensions including `DimModelConfig`, `DimClusterProfiling`, `DimDate`, `DimCustomerProfiling`.

3.1 Context

Write the project background, business context, and objectives here.

3.1.1 Main issue

Describe the key problem statements, current bottlenecks, and target solutions here.

Chapter 4

Power BI

4.1 Context

Write the project background, business context, and objectives here.

4.1.1 Main issue

Describe the key problem statements, current bottlenecks, and target solutions here.